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Editor-in-Chief
Computer Networks

Dear Editor,

We are pleased to submit our manuscript entitled "NEURA: Local Excitable Routing Control for Stress-Adaptive Networks" for consideration as a research article in *Computer Networks*.

The manuscript studies adaptive routing under localized stress from a control-plane perspective. Instead of optimizing only delivery or path quality, it examines how localized stress can be amplified into excessive control dissemination, route churn, and recovery-period oscillation. NEURA identifies and realizes a low-amplification routing-control operating point under localized network stress.

NEURA is proposed as a white-box local excitable routing-control law over a distance-vector-like route-quality advertisement substrate. It organizes integration, thresholded sparse signaling, refractory suppression, short-term memory, and guarded switching into an explicit mechanism for limiting routing-control amplification. The paper evaluates this operating point against periodic traffic engineering, triggered traffic engineering, ECMP-based traffic engineering, and lightweight learning-based routing using discrete-event simulation, mechanism attribution, sensitivity analysis, safety indicators, and packet-level ns-3 validation.

We believe the manuscript fits the scope of *Computer Networks* because it addresses a concrete network-control problem and contributes a complete distributed routing-control mechanism with a systematic evidence chain. The central claim is intentionally bounded: NEURA is not presented as a universal replacement for traffic engineering or as a new routing theory, but as a low-control, low-churn, usable-delivery operating point for resource-constrained stress-adaptive networks.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission. The authors declare no competing interests.

Thank you for considering our manuscript. We would be grateful for the opportunity to have it reviewed for publication in *Computer Networks*.

Sincerely,

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